

SAT Hard Question 15

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Novel Prep

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Excellence in SAT Prep

Question: Quadratic Function

Which of the following ensures that the Quadratic equation

$$ax^2 - 4x + c = 0$$

has at least one real root?

A $a > 0$

B $a = 0$

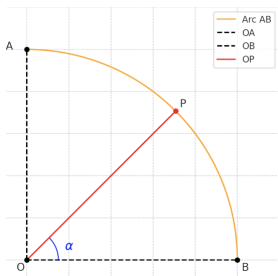
C $c > 0$

D $c = 0$

Answer

Answer: D

Question



As shown in the figure, O is the center of a circle with radius 1, and the circle intersects the coordinate axes at points A and B . Point P lies on arc $\widehat{AB}$ (not overlapping with A or B). Connect OP , and let $\angle POB = \alpha$. What are the coordinates of point P ?

- A $(\sin \alpha, \sin \alpha)$
- B $(\cos \alpha, \cos \alpha)$
- C $(\cos \alpha, \sin \alpha)$
- D $(\sin \alpha, \cos \alpha)$

Answer

Answer: C

Question: Stats

The following table shows the age distribution of a school choir's members:

Age (years)	13	14	15	16
Frequency	5	15	x	$10 - x$

For different values of x , which pair of statistical measures regarding age will not change?

- A Mean, Median
- B Mode, Median
- C Mean, Standard Deviation
- D Median, Standard Deviation

Answer

Answer: B

Question

Given that the line $y = kx + k + 1$ passes through the points $(m, n + 3)$ and $(m + 1, 2n - 1)$, and that $0 < k < 2$, what is a possible value of n ?

- A 3
- B 4
- C 5
- D 6

Question

The graph of the quadratic function $y = |a|x^2 + bx + c$ passes through the following five points:

$$A(m, n), \quad B(0, y_1), \quad C(3 - m, n), \quad D(\sqrt{2}, y_2), \quad E(2, y_3)$$

What is the correct order of y_1, y_2, y_3 ?

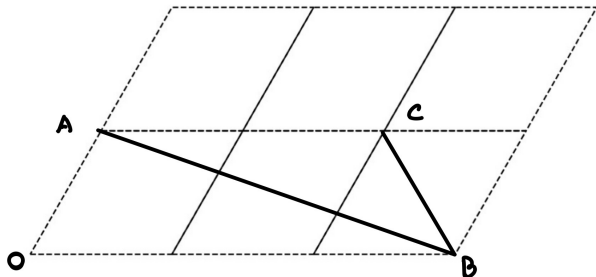
A $y_1 < y_2 < y_3$

B $y_1 < y_3 < y_2$

C $y_3 < y_2 < y_1$

D $y_2 < y_3 < y_1$

Question



As shown in the figure, 6 identical rhombuses form a grid. The vertices of the rhombuses are called grid points. It is known that one angle of the rhombus, $\angle O$, is 60° , and points A , B , and C are located on grid points. Find the value of $\tan \angle ABC$.

Question 10: Quadratic Function Reasoning

Given the quadratic function

$$y = x^2 - 2ax + a \quad (a \neq 0)$$

passes through two points

$$A\left(\frac{a}{2}, y_1\right), \quad B(3a, y_2),$$

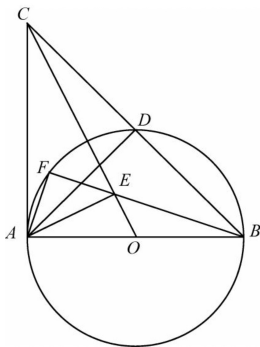
which of the following statements is correct?

- ☐ A. There exists a real number a such that $y_1 > a$.
- ☐ B. For all real numbers a , $y_1 > a$ is always true.
- ☐ C. There exists a real number a such that $y_2 < 0$.
- ☐ D. For all real numbers a , $y_2 < 0$ is always true.

Question

As shown in the figure, in triangle $\triangle ABC$, $\angle BAC = 90^\circ$, $AB = AC$, Circle O has diameter AB , and intersects BC at point D . Line $AE \perp OC$, with foot of the perpendicular at point E . The extension of line BE intersects AD at point F .

Find the value of $\frac{OE}{AE}$.



Answer

1/2

Question

In rectangle $ABCD$, point B is the center of a circle $\odot B$ with radius BC . The circle intersects AB at point F . Point E lies on AD , and segment CE is rotated about E to form EG such that point G lies on $\odot B$, where $\angle GEC = 90^\circ$. Point F is the midpoint of EG . Given that $AF = 1$, $AE = 3$, find the length of $CD =$ _____.

